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Incipient Fault Diagnosis in Ultra-Reliable Electrical Machines

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***Abstract*—Early detection of incipient faults in AC drives is one of the most difficult challenges for condition monitoring, especially for faults related to insulation degradation of the windings. In low power motors, the insulation degradation is principally due to the steep voltage variations caused by the voltage source converters (VSCs) that drive the machines. In the last years, with the coming to market of new no-Si based power devices, which achieve great values of dv/dt, fast tracking of electrical fault has became a topic of primary importance. In this paper a new method to detect the presence of incipient faults in ultra-reliable electric machines is presented and compared with a previous solution. The different methods were extensively evaluated, by means of experimental results. It is shown that potential winding faults can be detected at an early stage of fault inception and thus measures can be taken to limit propagation.**

Keywords*—DC-AC power converters, Electrical accidents, Electric machines, Electric motors, Fault detection, Fault diagnosis, Motor drives, Permanent magnet motors, Pulse width modulation converters, Variable speed drives

I. Introduction

In recent years, the rush for increased power density is pushing forward the use of high speed direct drive electrical motors driven by power converters based on wide-bandgap electronic switches (SiC and GaN devices). The pulse width modulated voltage, produced by these devices, consists of surges having rise times around 10 ns and commutation frequencies up to 100 kHz. Voltage surges travel on cables and, when they reach the motor terminals, due to impedance mismatch, they are reflected back to the converter, causing over-voltages at the machine terminals than can reach two or even three times the DC bus voltage. The problem was already known for motors fed by Si-based power converters, but it was a concern only for applications where the converter was placed at sufficiently long distance from the motor [1], [2]. With wide-bandgap based converters, also cables of less than 2 meters in length represent an issue and special attention must be payed in the electrical insulation motor design to prevent failures. In fact, although the advantages in employed wide-bandgap devices are mostly universally recognized in term of efficiency and power density, their impact on the reliability, and in particular on the insulation layer of electrical motors, has still to be fully assessed. Many researches are in progress to investigate the influence of short commutation times on electric drives lifetime, and it is not excluded that new design standards for electrical motors will be defined [3].

Over-voltages, triggered by fast dv/dt, can lead to the inception of partial discharges (PD) between adjacent turns. Under these conditions, the lifetime of the insulation might be extremely short: using PWM inverters with a commutation frequency of 10 kHz, failure times down to about 50 hours were observed in [4].

In applications where reliability is a major concern the required safety levels can be achieved in two ways. One approach is to design the machine such that it can tolerate faults; the other is to design the machine in such a way that the likelihood of faults occurring is reduced to an acceptable level. The latter aims to reduce the probability of failure achieving operating conditions at less stressful levels, as well as to implement prognostics and health monitoring techniques to prevent the occurrence of a severe failure.

Accordingly to this principle an alternative solution for the insulation system of electrical motors has been proposed in [5], where a 36 slot/6 pole machine with distributed windings used for an aircraft starter-generator was equipped with high voltage cables. The same concept was already adopted for high voltage power generators and oil free transformers: PowerformerTM by ABB and Alstom Power [6], WindformerTM and DryformerTM both by ABB [7].

In [8] the insulation system is very similar to the insulation system of a polymer cable (see Fig. 1). Between the circular conductor and the polymer insulation, there is a semiconducting layer, also known as semicon layer, to eliminate locally high electric fields. For the same reason, there is a semiconducting layer outside the insulation. However, in contrast to a cable used for transmission, this layer is not continuously grounded by an outer conductive screen, but each turn is connected to the stator core in a single point by means of silver paint and copper foil. Conversely, in [5] the semicon layer is covered by an additional thin insulation layer to prevent direct electrical connection with the stator core.

An important difference between this new insulation system and the conventional one, which makes it really suitable for wide-bandgap converters, is that the electrical stress between the turns in the new system is strongly reduced. In fact, the outer semiconducting layer screens the electric coupling between the turns of the winding, thus the turn-to-

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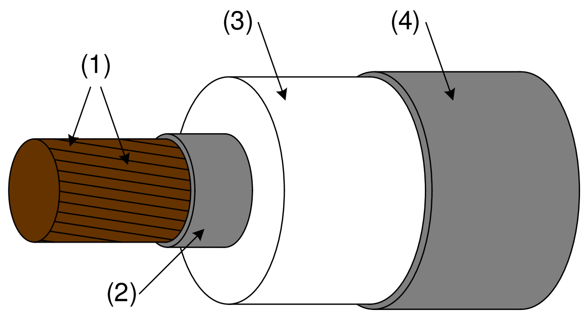

Fig. 1. Conductor of the insulation system of the rotating electric machine studied in [8] : (1) a stranded conductor, (2) an inner semiconducting layer, (3) an insulator, and (4) an outer semiconducting layer.

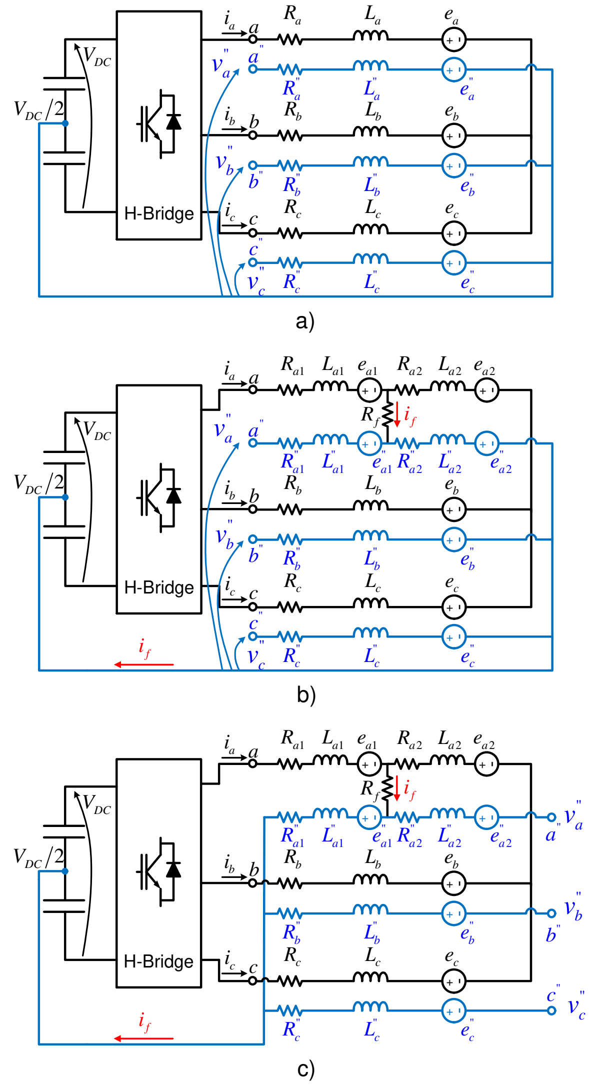

Fig. 2. Stator winding connections diagram of the healthy (a) and faulty (b) machine. In (c) is shown the case of faulty motor when the windings are connected in a different way.

turn short circuit failure probability is reduced. Nevertheless, the capacitive current through the insulation of the cable and the eddy currents in a conductor, which are induced by the magnetic fields inside it, cause additional losses and heating. In [9] a lumped circuit to model the capacitive current and the losses in the outer semicon of a turn in a coil was developed. It shows as the semicon resistance has an important role for the damping in the coil and reducing losses. Before a turn-to-turn fault occurs, a leakage current through the semicon layer, due to degradation of the intermediate insulation layer, will occur. Sensing this current could give reliable information to detect incipient faults in early stage, avoiding subsequent damages and enhancing the safety level of the machine.

Different monitoring techniques for fast detection of electrical faults have been suggested in literature for electrical drives [10]–[13], but for the particular structure proposed in [5] novel solutions are proposed in this paper; advantages and disadvantages of each method are here compared and evaluated throughout simulations and experimental results. This work was previously presented in [14] and it is reproposed and revised.

II. PMSM Motor Model

Although several models and mathematical studies have been elaborated in order to describe the behavior of faulty and healthy PMSM motors [15], due to the particular insulation system of the studied motor, a custom model of the machine has to be developed. A low frequency model for healthy and faulty machine was originally developed in [16] and it is reported in Fig. 2 along with the connection scheme of the drive. In a coil the inner conductive layer and the outer semicon are exposed to the same flux linkage, thus for each turn the induced voltage in the conductor is nearly the same as the voltage in the outer semicon layer. Considering only low frequency components it is possible to simplify the analysis by neglecting the capacitive coupling between the conductor and semicon layer. In this case, the inner conductive and outer semicon layers can be conceived as two electrically separated cables, wound around the same coil with equal number of turns. In this way, the motor can be considered as having two separate three-phase windings. One three-phase winding, represented by phases a, b, c in black in Fig. 2, is due to the inner conductors of the cables, and it is connected to the Voltage Source Converter (VSC) as usual. Therefore, the a, b, c phases are here referred as main windings. The second three-phase winding is obtained from the semicon layers of the same cables. It is represented by the blue a'', b'', c'' phases in Fig. 2, referred in this paper as search windings, as they are used for monitoring purposes. The main difference between the two three-phase windings is that the neutral point of the three-phase system related to the semicon layers is connected to the middle point of the DC-link, whereas the other is floating. Despite of that, it can be expected that main and search windings present the same value of self-inductance and the same flux linkage, i.e. the same back emf. Furthermore, the magnetic coupling factor between them is close to the unity as the small leakage flux inside the cable itself is not considered in this analysis. Hence the system equation can be written as in (1) where the following approximation are made: R_a=R_b=R_c=R, R''_a=R''_b=R''_c=R'', L_a=L_b=L_c=L, L_a=L''_a=$M_{aa''}$=L, $M_{ab''}$=$M_{ac''}$=$M_{ba''}$=$M_{bc''}$=$M_{cb''}$=$M_{ca''}$=M and e_a=e''_a,

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e_b=e''_b, e_c=e''_c.

$$[V_s] = [R_s] \cdot [i_s] + [L_s] \cdot \frac{d}{dt}[i_s] + [e_s] \tag{1}$$

$[V_s]$ takes into account both the voltages of the main windings $[V_{mw}]$=$[V_a, V_b, V_c]$, and the voltages of the search windings $[v_{sw}]$=$[v''_a, v''_b, v''_c]$, being $[V_s]$=$[[V_{mw}], [v_{sw}]]^t$. In the same way $[i_s]$=$[[i_{mw}], [i_{sw}]]^t$=$[i_a, i_b, i_c], [i''_a, i''_b, i''_c]]^t$, and $[e_s]$=$[[e_{mw}], [e_{sw}]]^t$ with $[e_{mw}]$=$[e_{sw}]$=$[e_a, e_b, e_c]$. Moreover, the simplified resistance and inductance matrices are shown in (2):

$$[R_s] = \begin{bmatrix} [R] & 0 \\ 0 & [R''] \end{bmatrix}, [L_s] = \begin{bmatrix} [L] & [L] \\ [L] & [L] \end{bmatrix} \tag{2}$$

where:

$$\begin{gathered} [R] = \begin{bmatrix} R & 0 & 0 \\ 0 & R & 0 \\ 0 & 0 & R \end{bmatrix}, \quad [R''] = \begin{bmatrix} R'' & 0 & 0 \\ 0 & R'' & 0 \\ 0 & 0 & R'' \end{bmatrix}, \\ [L] = \begin{bmatrix} L & M & M \\ M & L & M \\ M & M & L \end{bmatrix} \end{gathered} \tag{3}$$

In case of healthy motor, the search windings currents i''_a, i''_b, i''_c are equal to zero and the system equation can be further simplified as in (4).

$$\begin{cases} [V_{mw}] = [R_s] \cdot [i_s] + [L_s] \cdot \frac{d}{dt}[i_s] + [e_{mw}] \\ [v_{sw}] = [L_s] \cdot \frac{d}{dt}[i_s] + [e_{sw}] \\ [e_{sw}] = [e_{mw}] \end{cases} \tag{4}$$

An incipient fault, which involves a degradation of the insulation layer of the cable, can be modeled as resistance between the conductor and semicon layer, therefore between a main and a search winding in the model. Fig. 2b depicts the motor in case of an incipient fault between phases a and a''. In this figure, the windings of phases a and a'' are described with two sub-windings in series a_1, a_2 for a and a''_1, a''_2 for a'' respectively. The separation point of the two sub-windings represents the turn where the insulation layer between the inner conductor and the outer semicon layer of the cable is degraded. The resistance R_f is used to model the damage in the isolation layer. Its value depends on the fault severity, when R_f decreases toward zero the fault evolves toward a full short circuit. In Fig. 2b, R_{a1}, L_{a1} and e_{a1} are the resistance, self-inductance and the back EMF of sub-winding a_1, whereas R_{a2}, L_{a2} and e_{a2} are the corresponding parameters of sub-winding a_2. The same concept is applied to the sub-windings of a''. Considering a total number N of turns for each winding and a damage in the insulation layer in correspondence of the n turn, it is possible to express the system parameters in function of the ratio μ=n/N. Equation (5) reports the dependence of the system parameters respect to the fault position; the mutual inductance M_{a1a2} between sub-windings a_1 and a_2 it is also calculated.

$$\begin{cases} L_{a1} = L''_{a1} = (1-\mu)^2 L, L_{a2} = L''_{a2} = \mu^2 L \\ M_{a1a2} = M_{a1''a2''} = \mu(1-\mu)L \\ M_{a1a1''} = L_{a1}, M_{a2''a2''} = L_{a2} \\ M_{a1b} = M_{a1''b} = M_{a1c} = M_{a1''c} = (1-\mu)M \\ M_{a2b} = M_{a2''b} = M_{a2c} = M_{a2''c} = \mu M \\ R_{a1} = (1-\mu)R, R_{a2} = \mu R \\ R''_{a1} = (1-\mu)R'', R''_{a2} = \mu R'' \\ e_{a1} = e''_{a1} = (1-\mu)e_a, e_{a2} = e''_{a2} = \mu e_a \end{cases} \tag{5}$$

The analysis considers a concentrated-windings machine, but it is still a good approximation in case of distributed-windings. The total voltage of the phase a (V_a) is given by the sum of the voltages across the sub-windings a_1 and a_2, V_a=v_{a1}+v_{a2}. The same consideration is valid for a'' with v''_a=$v_{a1''}$+$v_{a2''}$. Moreover, indicating with i_f the fault current as in Fig. 2b, from the circuit analysis it is possible to obtain the equations in (6).

$$\begin{cases} V_a = v_{a1} + v_{a2} \\ v''_a = v_{a1''} + v_{a2''} \\ i_{a1} = i_a \\ i_{a2} = i_{a1} - i_f \\ i_{a2''} = i_f \end{cases} \tag{6}$$

Integrating (6) in (1) leads to the system equation for the faulty motor in (7). Replacing (5) in (7), the simplified system equation, that express the voltages at main and search windings terminals for the faulty motor ($[V_{sf}]$) is given in (8):

$$[V_{sf}] = [R_{sf}] \cdot [i_{sf}] + [L_{sf}] \cdot \frac{d}{dt}[i_{sf}] + [e_{sf}] \tag{8}$$

being $[V_{sf}]$=$[[V_{mw}], [v_{sw}]]^t$, $[i_{sf}]$=$[i_a, i_b, i_c, i_f]^t$=, $[e_{sf}]$=$[[e_{mw}], [e_{sw}]]^t$, and the matrices of resistances and inductances as in (9).

$$\begin{gathered} [R_{sf}] = \begin{bmatrix} R & 0 & 0 & -\mu R \\ 0 & R & 0 & 0 \\ 0 & 0 & R & 0 \\ 0 & 0 & 0 & -\mu R'' \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \\ [L_{sf}] = \begin{bmatrix} L & M & M & 0 \\ M & L & M & 0 \\ M & M & L & 0 \\ L & M & M & 0 \\ M & L & M & 0 \\ M & M & L & 0 \end{bmatrix} \end{gathered} \tag{9}$$

As can be seen, in the inductance matrix no variation is induced by the presence of a fault current i_f, flowing through the search winding a''. Only a slight variation in the resistance matrix is present, with a strong dependence on the fault position (μ) and on the amplitude of the fault current. Nevertheless, in case of an incipient fault at the very beginning state, the amplitude of i_f is expected to be low, as the insulation degradation has not yet evolved in a short-circuit (high R_f), thus even the variation of the resistance matrix is negligible. Therefore, it is not possible to detect the presence of a fault simply monitoring the amplitude of fundamental of the main/search windings voltages $[V_{sf}]$. Moreover, since the inductance matrix

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$$\begin{bmatrix} v_{a1} \\ v_{a2} \\ V_b \\ V_c \\ v''_{a1} \\ v''_{a2} \\ v''_b \\ v''_c \end{bmatrix} = [1]_{8x8} \cdot \begin{bmatrix} R_{a1} \\ R_{a1} \\ R \\ R \\ R''_{a1} \\ R''_{a1} \\ R''_b \\ R''_c \end{bmatrix} \cdot \begin{bmatrix} i_a \\ i_a - i_f \\ i_b \\ i_c \\ 0 \\ i_f \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} L_{a1} & M_{a1a2} & M_{a1b} & M_{a1c} & M_{a1a1''} & M_{a1a2''} & M_{a1b''} & M_{a1c''} \\ M_{a1a2} & L_{a2} & M_{a2b} & M_{a2c} & M_{a2a1''} & M_{a2a2''} & M_{a2b''} & M_{a2c''} \\ M_{a1b} & M_{a2b} & L & M & M_{ba1''} & M_{ba2''} & L & M \\ M_{a1c} & M_{a2c} & M & L & M_{ca1''} & M_{ca2''} & M & L \\ M_{a1a1''} & M_{a2a1''} & M_{ba1''} & M_{ca1''} & L_{a1} & M_{a1a2} & M_{a1b''} & M_{a1c''} \\ M_{a1a2''} & M_{a2a2''} & M_{ba2''} & M_{ca2''} & M_{a1a2} & L_{a2} & M_{a2b''} & M_{a2c''} \\ M_{a1b''} & M_{a2b''} & L & M & M_{a1''b''} & M_{a2''b''} & L & M \\ M_{a1c''} & M_{a2c''} & M & L & M_{a1''c''} & M_{a2''c''} & M & L \end{bmatrix} \cdot \frac{d}{dt} \begin{bmatrix} i_a \\ i_a - i_f \\ i_b \\ i_c \\ 0 \\ i_f \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} e_{a1} \\ e_{a2} \\ e_b \\ e_c \\ e''_{a1} \\ e''_{a2} \\ e''_b \\ e''_c \end{bmatrix} \tag{7}$$

does not vary, diagnostic systems that monitor changes in the magnetic behavior of the electric machine, such us high frequency current injection technique, are not feasible in this case. Nevertheless, it must be considered that the harmonic content of the back EMF voltages and the supply voltages $[V_{mw}]$ is not limited only to the fundamental one. In actual implementations back EMF presents third, fifth, and other multiple order harmonics. Even if design guidelines for electric motor lead to a minimization of the flux link harmonics, a pure sinusoidal waveform remains difficult to obtain. Furthermore, the most used PWM modulation technique for electric drive is the Space Vector Modulation (SVM). SVM is preferred to sinusoidal PWM because the generated output voltages have an amplitude up to 15% greater than in the case of sinusoidal PWM with same value of input DC voltage. This behavior is possible because SVM intrinsically achieves over-modulation, adding a third harmonic in the generated phase voltages. In a three-phase system, as the case of an electric motor, the third harmonic is canceled in the line-to-line voltages [17]. However, if an incipient fault is present the third harmonic of the supply phase voltages causes the current i_f having a third harmonic as well. Therefore, it is possible to express the fault current i_f as in (10).

$$\begin{cases} R_f i_f = v_{a2} - v_{a2''} + (v_{sp} - v''_{sp}) \\ i_f = i_a - i_{a2} \\ i_{a2} + i_b + i_c = 0 \end{cases} \tag{10}$$

It is possible to individuate two components for the current i_f. One component is due to the fundamental of the supply voltage (V_a for the example in Fig. 2b) and, therefore, to the first harmonic of the supply frequency. The other component is due to the third harmonics of V_a and to the difference between the neutral point voltages of the two three-phase systems. As widely investigated, in presence of a third harmonic component in the back EMF or in the supply voltages of the motor, the neutral point voltage in a wye-connected stator present an oscillation at that frequency. Since the neutral point of the search phases (v''_{sp}) is clamped to a fixed voltage value (i.e. the midpoint voltage of the DC-Link as in Fig. 2), the fault current will present a third harmonic component due to difference $v_{sp} - v''_{sp}$.

A. Proof of Concept Simulations

The model for healthy and faulty PMSM motor was simulated using PLECS toolbox in Simulink/Matlab environment. Simulations were performed with the attempt to verify the analysis in II. Each winding was divided in two sub-windings, for reproducing the case of Fig 2b. The parameters of each sub-winding are constituted by a fraction of the total phase resistance, phase inductance and back EMF of the motor. The total parameters of the PMSM machine are reported in Table I. Since the same number of turns was chosen for the sub-windings, they were considered identical, and the values adopted for phase resistances R, the self-inductances L and mutual inductances M were $R = R'' = 0.94/2\Omega$, $L = L'' = 2.42/2mH$, and $M = M'' = -1.21/2mH$. Although the resistance of the semicon material is higher than the conductor's one, the same value was considered in the simulation for being in accordance with the experimental setup. Furthermore, the harmonic content of the back EMF was included in the model, considering a value of third harmonic equal to one tenth of the fundamental one. The square wave generators v_a, v_b, v_c represent the output voltages of the power converter employed to drive the motor. Since simulations act as proof of concept, only the low frequency components (i.e. sinusoidal fundamental and multiples of it) of the inverter output voltages are included in the model. They synthesize the voltage waveforms that are obtained with Space Vector Modulation SVM technique. The resistor R_f was used to simulate a damage in the dielectric layer of the cable. Fig. 3 shows V_{mw}, v_{sw} and i_{mw} in case of healthy motor. In the Figure it is clearly recognizable the low frequency shape waveform of SVM modulated inverter output voltage. SVM intrinsically adds a third harmonic in the generated phase voltages that is canceled in the line-to-line voltages. As can be seen i_f is equal to zero and the phase currents and voltages are not distorted. Fig. 4 shows the same waveforms in case of a fault occurrence between inner conductor and outer semicon in Fig. 2b when R_f is equal to 10Ω. The third harmonic content of i_f, due to the third harmonics of V_a and to the difference between the neutral point voltages of the two three-phase systems, is clearly notable and superimposed to the fundamental one. It is also noticeable that, although no-distortion is appreciable in V_{mw} and v_{sw} waveforms, the currents i_{mw} result unbalanced.

III. Comparison of Different Diagnostic Methods

As described before, in case of an incipient fault in the insulation layer of the cable a third harmonic component arises in the supply currents. The third harmonic is due to the fault current i_f that flows through the fault resistance R_f toward

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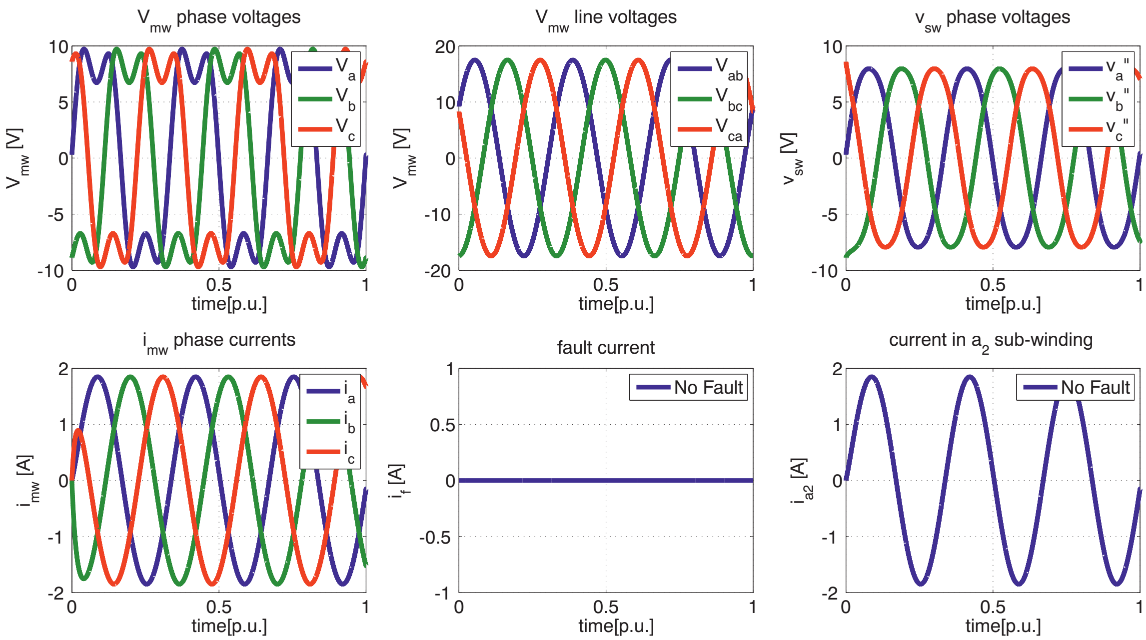

Fig. 3. Simulation Results V_{mw}, v_{sw}, i_{mw}, i_f and i_{a2} in case of No fault. Time is in per unit of 42.9 ms.

the mid-point of the supply voltage, to which the neutral point of the search windings is connected. The amplitude of the third harmonic is much lower than the fundamental one and it is related to the load condition, the rotating speed, and to the severity of the fault. Two different methods to detect the presence of an incipient fault are described in the subsequent sub-paragraphs.

A. Analysis of the Harmonic Content of the Instantaneous Active Power

In [16] the supply currents of the main windings and the voltages of the search windings are sampled and computed to obtain the instantaneous active and reactive power absorption of the motor in $\alpha\beta0$ reference frame. Since the proposed methods focus on low harmonics multiple of the fundamental frequency, the analysis is independent of the the high frequency components introduced by PWM modulation, such as current or voltage ripples, which, according to the modulation technique adopted for the inverter, could be at the same frequency of the switching carrier or multiples of it. The sampled voltages and currents are filtered in order to acquired only low frequency components, thus the amplitude of current and voltage ripples does not affect the feasibility of the proposed methods. The instantaneous active power thus calculated in the $\alpha\beta0$ reference frame presents an average part ($\bar{p}$) and an alternating part ($\tilde{p}$) in (11). In particular, the fourth harmonic of the active power is given by the cross product of the fundamental of the voltage with the third harmonic of the current [16]. This characteristic suggests that the fourth harmonic of the active power will be present only in case of an incipient fault in the insulation layer of the cable. Moreover, even if the amplitude of the third harmonic of the current is low, the fourth harmonic of the instantaneous powers should be significant because of the contribution of the fundamental of the supply voltage. The presence of a fault is then detected by means the integral of the active power in a reference frame synchronous with the forth harmonic of the supply frequency. In fact, in a rotating reference frame, only the space-vector that is synchronous with the frame has a constant value, whereas the space-vectors of all the other harmonics have pulsating components with zero mean value (Fig. 5).

$$p = \bar{p} + \widetilde{p_{2\omega}} + \widetilde{p_{4\omega}} + \widetilde{p_{6\omega}} \tag{11}$$

This method allows detecting incipient faults but it suffers from a strong dependence on the load conditions, the rotating speed of the machine and the position of the fault. In fact, if the fault occurs near the phase terminals the fault current amplitude is greater than the case of a fault occurrence near the neutral point of the windings. In particular, if the degradation of the insulation layer occurs in correspondence of the neutral point, the presence of the fault is not detected at all. This drawback is partially mitigated by the fact that the degradation of the insulation layer is more likely expected near the terminals of the windings. In fact, the electrical stress on the insulation layer, due to the PMW strategy adopted by VSCs, is not uniformly distributed along the windings, but it is more prominent in the first turns of the windings. Moreover, the method proposed in [16] employs three current sensors and three voltage sensors. Even if just two current sensors are sufficient for driving a motor, in reliable applications three current sensors are usually embedded in the VCS, for enabling alternative control strategies in case of breakdown of one phase of the drive. Therefore, no additional current sensors are required for monitoring incipient faults. On the contrary, the voltage sensors for the search windings are additional respect to a conventional VSC.

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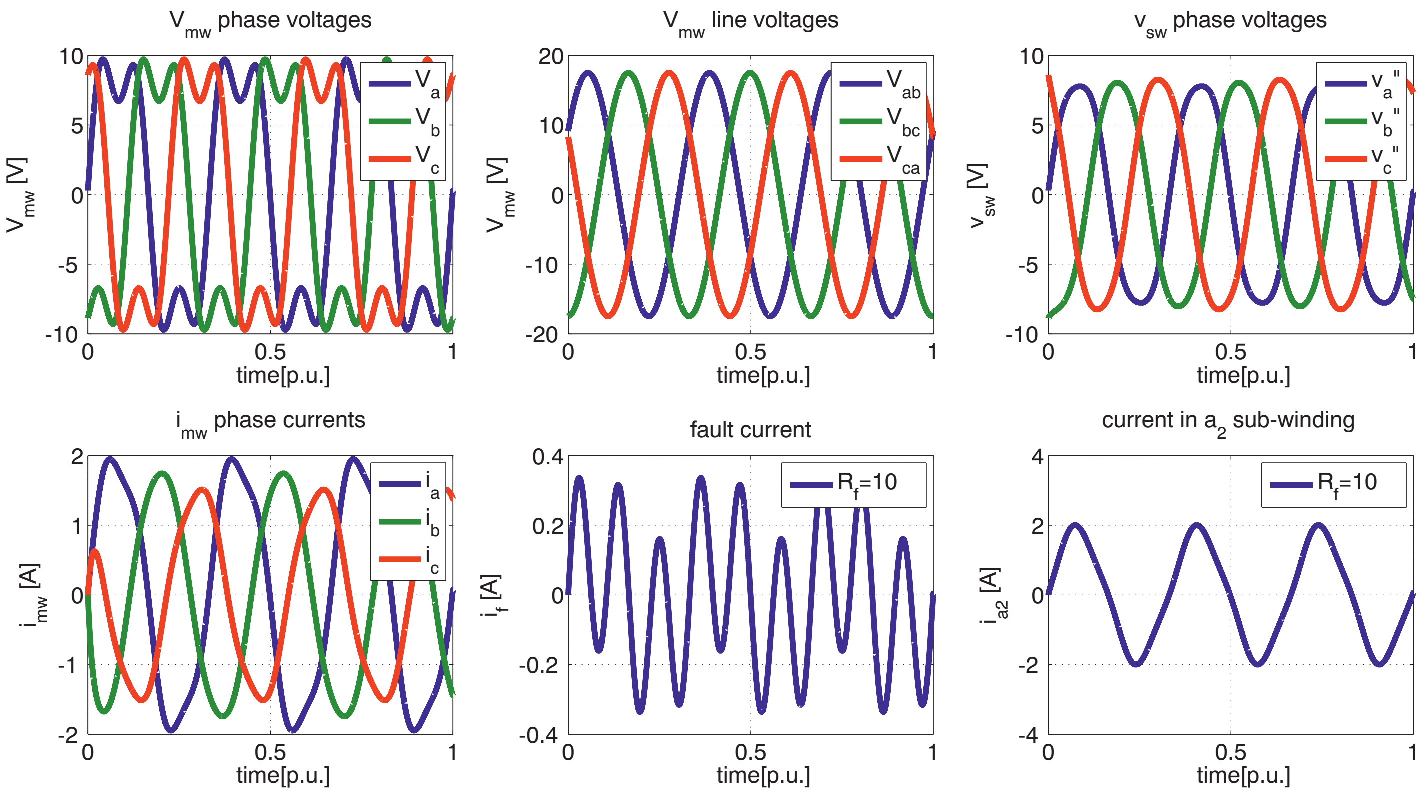

Fig. 4. Simulation Results V_{mw}, v_{sw}, i_{mw}, i_f and i_{a2} in case of a fault between points $2-2'$ in Fig. 7 and $R_f = 10\Omega$. Time is in per unit of 42.9 ms.

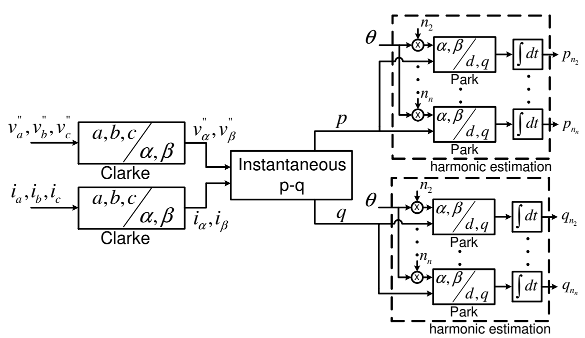

Fig. 5. Fault Control Process.

B. Sensing the Current in the connection to the DC link mid-point

A simple alternative to the method proposed in [16] could be to add a current sensor in the path that connects the neutral point of the search windings to the mid-point of the DC source. As explained in the previous section, the current that flows in the fault resistance is the same that circulates in the connection between the neutral point of the search winding and the mid-point of the DC-link. This method has the advantage to allow the detection of i_f even when the fault occurs at the neutral point of the windings. Due to the low amplitude of i_f respect to the supply currents of the motor, the current sensor employed for sensing i_f cannot be the same as those used for the main windings; in fact it must present a high sensitivity. The commercial current sensors that can detect a current with amplitude of few mA do not usually present an high value of maximum current rating, thus a possible damage of the sensor in case the incipient fault would evolve in a severe short-circuit has to be taken into account. A possible solution could be to rearrange the phase connections in order to change the position of the terminals of the search windings as in Fig. 2c. In this way the positions of the terminals of the main and search windings are inverted and the differential voltage between them is higher than in the previous case. Consequently, for a given value of R_f the current i_f is greater and a current sensor with higher maximum current rating can be used to detect the presence of a fault. Nevertheless, the electrical stress on the insulation layer of the cable is greater as well, and a decreasing of the motor lifetime is to be expected. Despite of the winding connection, the proposed method adopts one additional current sensor respect to the case of a conventional VSC, and no additional voltage sensors. Therefore, it presents a lower impact in term of costs respect to the solution in [16]. Moreover, the complexity of the control is reduced, since no calculations are needed but just a comparator is sufficient to easily detect the presence of an incipient fault. The performances of the two solutions were evaluated throughout simulations and experimental results.

IV. Experimental Results

The experimental tests were performed with a prototype motor whose electric parameters are shown in Table I. Since a PMSM with the same insulation system as in [5] was not available, an additional cable (the blue one in Fig. 6) was added in each phase in order to simulate the presence of the semicon layer of the cable. This represents a good approximation of the system at low frequency. Several point

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TABLE I. ELECTRICAL PARAMETERS OF THE PMSM

Number of Poles	28
Nominal Speed	$5000rpm$
Nominal Torque	$15Nm$
Nominal Power	$7.85kW$
K_t	$1.6Nm/A$
K_e	$1.33V_{rms}/krpm$
Phase Resistance	0.94Ω
Phase Inductance	$2.42mH$

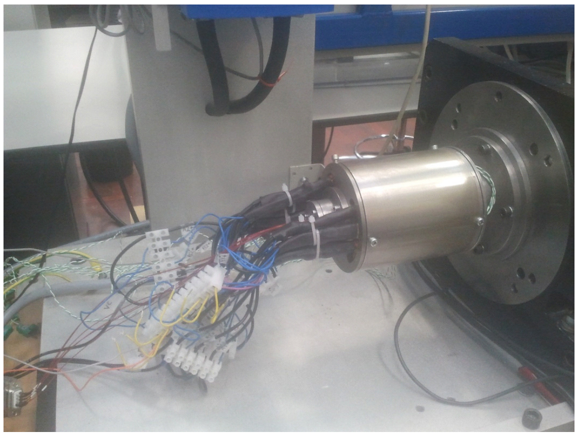

Fig. 6. Prototype motor used for the experimental test.

of access to the windings were provided to reproduce different fault positions and a schematic of the electric connection is show in Fig. 7. The fault resistance R_f was simulated with a variable resistor connected between a main winding and a search winding as in Fig. 7. Values of 100, 10, 1, 0.2Ω of R_f were considered to simulate different severity levels of the incipient fault. The variable resistor was disconnected when the behaviour of the healthy motor was reproduced. In order to prove the feasibility of the proposed diagnostic methods also in case of low speed and light load, the experimental result were performed considering a mechanical speed range from 180 to 780 rpm, and a torque of $1Nm$, as in the case of numerical simulations. These values are considerably lower than the nominal ones reported in Table I.

In Fig. 8a,b the calculated amplitude of the forth harmonic of the instantaneous active power in the $\alpha\beta0$ reference frame is reported for different values of R_f and mechanical speed ω_r (Fig. 8a), whereas Fig. 8b shows the same variable in function

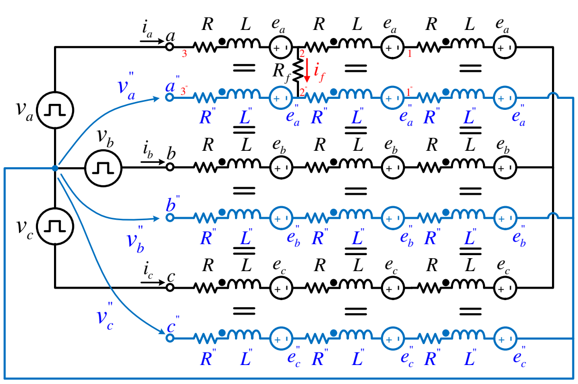

Fig. 7. Schematic of the motor connections for the experimental test.

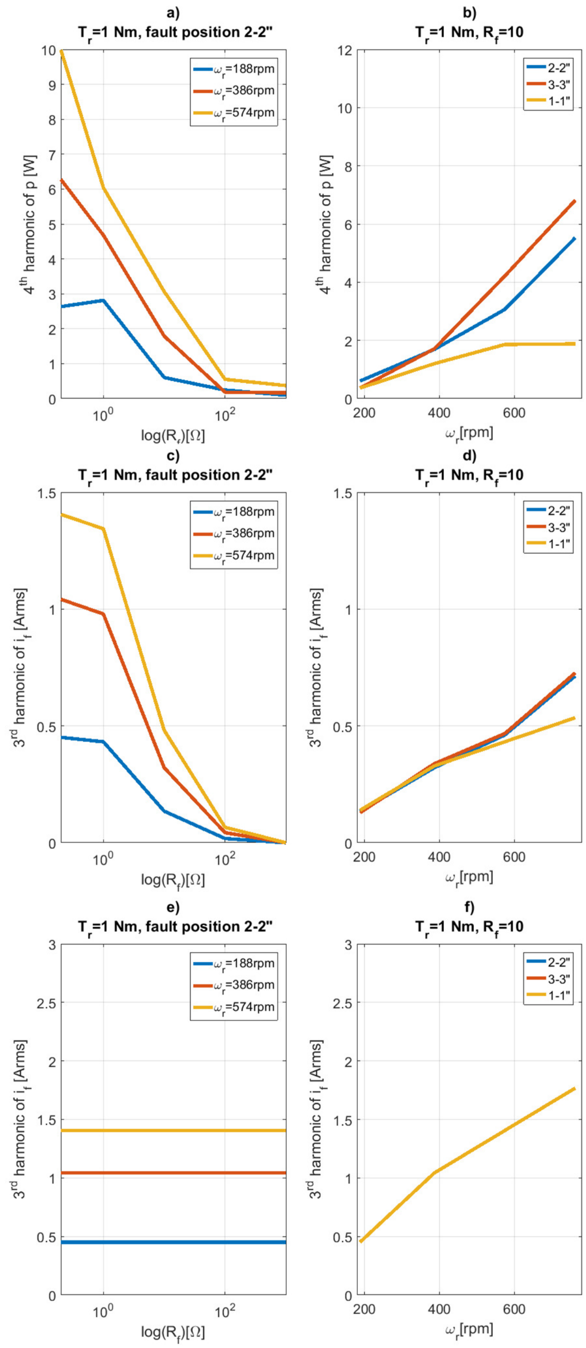

Fig. 8. Amplitude of the fourth harmonic of the active power (a) third harmonic of i_f (c) as a function of R_f, for different values of the mechanical speeds ω_r, and rated torque $T_r = 1Nm$. Amplitude of the fourth harmonic of the active power (b) third harmonic of if (d) as a function of ω_r, for different positions of the fault, $T_r = 1Nm$ and $R_f, = 10\Omega$. In (e) and (f) the amplitude of the third harmonic of i_f is plotted under the same test conditions as (c) and (d) but when the windings are connected as in Fig. 2c.

of ω_r when $R_f = 10\Omega$ and different positions for the incipient fault are considered ($1-1'$, $2-2'$ and $3-3'$ in Fig. 7). As can be seen the method proposed in [16] presents a dependence with the speed, the severity of the fault and the position of the fault. Furthermore, as in the simulations, two regions of operation can be individuated: when the value of R_f is high, the amplitude of forth harmonic of the active power is nearly

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constant, otherwise, when the severity of the fault increases (i.e R_f decreases), the amplitude starts to rise linearly. The value of R_f between the two states of behaviour is near 10Ω. Fig 8c,d show the amplitude of the third harmonic of the fault current, detected with the method proposed in this paper, in the same test conditions of Fig. 8a and Fig. 8b respectively. As expected, the fault can be detected even when it occurs in correspondence of the neutral point of the windings. Fig. 8e,f report the amplitude of the third harmonic of i_f under the same test conditions but when the windings are connected as in Fig. 2c. The current amplitude is greater than those in Fig. 8c,d as expected. Therefore, although it requires an additional current sensor respect to the strategy in [16], the proposed solution allows a more simple detection method for incipient faults. Moreover, since it does not present an insensibility for high value of R_f, it can effectively detect incipient faults even in their early stage. This represents the main advantage respect to the solution in [16].

V. Conclusion

In this paper a comparison of different methods for incipient fault detection in ultra-reliable PMSM is performed. The advantages and disadvantages of each method are investigated and analysed throughout theoretical considerations and experimental results. To this purpose a prototype motor for simulating the presence of incipient fault at different severity levels was built. The results show that the proposed methods can be effectively employed as on-line condition monitoring systems.

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